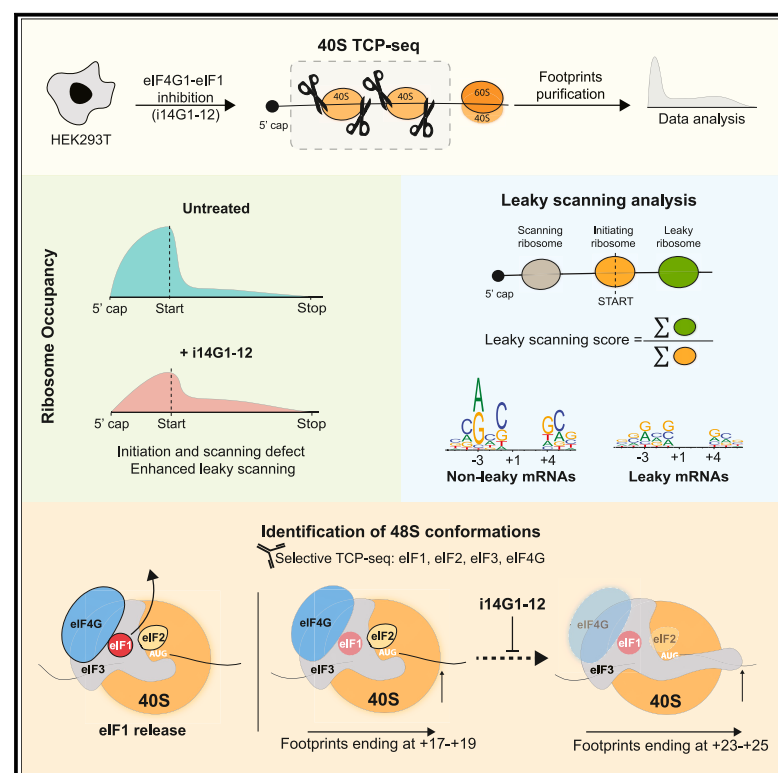


Unraveling the landscapes and regulation of scanning, leaky scanning, and 48S initiation complex conformations

Graphical abstract



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In brief

Weiss and Dikstein report that during translation initiation in eukaryotes the inhibition of eIF4G1-eIF1 leads to scanning defects and enhanced leaky scanning. They also identify multiple conformations of the 48S complex by analyzing their footprints *in vivo*. These are proposed to be sequential and have different factor compositions.

Highlights

- Studying scanning and initiating 48S and eIF4G1-eIF1 inhibition unveil important features
- A leaky scanning map uncovers eIF4G1-eIF1 and AUG context importance
- Two major conformations of the initiating 48S differentially engage with eIFs identified
- i14G1-12 inhibitory effect on eIF1-eIF4G1 is linked to eIF3



Article

Unraveling the landscapes and regulation of scanning, leaky scanning, and 48S initiation complex conformations

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SUMMARY

Scanning and initiation are critical steps in translation. Here, we utilized translation complex profiling (TCP-seq) to investigate 48S organization and eIF4G1-eIF1 inhibition impact. We provide global views of scanning and leaky scanning, uncovering a central role of eIF4G1-eIF1 in their regulation. We confirm AUG context importance, with non-leaky genes featuring a Kozak context and cytosine at positions -1 and $+5$. Capturing 48S complexes associated with eIF1, eIF4G1, eIF3, and eIF2 through selective TCP-seq revealed that the eIF3-scanning ribosome is highly vulnerable to eIF4G1-eIF1 inhibition, and eIF1 tends to dissociate upon AUG recognition. Initiation-site footprint analysis revealed a class spanning -12 to $+18/19$ from the AUG, representing the entire 48S and enriched with eIF2, eIF1, and eIF4G1, indicative of early initiation. Another eIF3-dependent class extends up to $+26$ and exhibits reduced eIF2 and eIF4G1 association, suggesting a late/alternative initiation complex. Our analysis provides an overview of scanning, initiation, and evidence for conformational rearrangements *in vivo*.

INTRODUCTION

Eukaryotic mRNA translation is carried out by an intricate machinery consisting of ribosomes, tRNAs, and multiple accessory factors. It is a cyclic process that includes four successive steps: initiation, elongation, termination, and subunit recycling, of which initiation is the most regulated stage.¹ The translation initiation phase involves the loading of the small ribosome subunit (40S) onto the mRNA at the 5' end, followed by scanning of the 5' untranslated region (UTR; or leader sequence) and start codon (AUG) recognition. During this process, the ribosome is assisted by a plethora of eukaryotic translation initiation factors (eIFs).² According to the canonical model of eukaryotic translation initiation, the 40S first assembles into the 43S preinitiation complex (PIC) with the multiprotein complex eIF3, the ternary complex eIF2-bound guanosine triphosphate (GTP) and tRNA^{Met}, eIF1/1A, and eIF5.^{3,4} The PIC is then recruited by the eIF4F complex sitting on the mRNA 5' m7G cap.⁵ eIF4F comprises the scaffolding protein eIF4G1, the cap-binding protein eIF4E, and the helicase eIF4A.² Subsequently, the PIC scans the leader sequence until it finds a start codon. The recognition of the latter triggers the release of phosphate from the hydrolysis of eIF2-bound GTP and of eIFs, followed by the recruitment of the large ribosome subunit (60S) to form the 80S ribosome.^{6–9}

Start codon recognition is the decisive step that arrests scanning and promotes the transition to elongation. It determines what open reading frame (ORF) is translated and, consequently, which protein sequence will be produced. Start codon recogni-

tion highly depends on the sequences flanking the AUG. A purine at -3 and a guanine at $+4$ were identified to be the most critical nucleotides to promote initiation and are defined as the Kozak context.^{10,11} The presence of a suboptimal AUG context may cause the 48S to bypass the start codon.¹² This process, known as leaky scanning, leads to initiation at downstream initiation sites and the production of truncated or different peptide sequences. Start codon recognition is regulated by eIF1 and eIF1A bound to the P and A sites of the ribosome, respectively.¹³ These factors distinguish the AUG context and are essential for the fidelity of the start site selection.^{6,14,15} Correct codon-anticodon base-pairing between tRNA^{Met} and the start codon induces a conformational change from an “open” to a “close,” resulting in eIF1 disengagement and phosphate release that prevents the ribosome from continuing scanning.^{9,13,16} While leaky scanning has been extensively studied using either reporter genes or specific mRNAs, the extent of this phenomenon *in vivo* and at a global scale and the features regulating it were not yet investigated.

In addition to its interaction with eIF3 in the PIC, eIF1 binds eIF4G1. This interaction, initially identified in yeast biochemically and genetically, is required to promote a stringent AUG selection.^{17–20} Their interaction was subsequently shown to be conserved in mammals and to inhibit initiation from cap-proximal AUG in a manner dependent on AUG context.^{21–23} Recently, the role of this interaction was studied by two small molecules, i14G1-10 and i14G1-12, identified through a high-throughput *in vitro* drug screen that is based on protein-protein interaction



and subsequent functional translation assays. While i14G1-10 mainly targets eIF4G1, i14G1-12 equally binds eIF4G1 and eIF1 but does not disrupt their interaction. Both chemically distinct molecules inhibit translation and cell growth via the same regulatory features.²³ However, presently, there is little insight into the impact of eIF1-eIF4G1 inhibition on scanning and initiation.

Besides the major involvement of eIF4G and eIF4A in translation initiation, a recent study showed that they preferentially retained on elongating ribosomes at upstream ORFs (uORFs).²⁴ Additionally, emerging evidence indicates that eIF3 remains in association with 80S ribosomes during the translation of short uORFs to promote re-initiation, and it can also remain associated with post-initiation 80S ribosomes to facilitate the early elongation of mRNAs.^{25–27}

In this study, we adapted the translation complex profiling (TCP-seq) and the selective TCP-seq (Sel-TCP-seq) methods^{28–30} in the context of eIF1-eIF4G1 inhibition by i14G1-12 to investigate the scanning and initiation steps by the 48S. Our analysis reveals that uORFs diminish ribosome accumulation and queuing in the vicinity of the start site. Additionally, we uncover the global landscape of leaky scanning and the critical role that eIF1-eIF4G1 plays in its restriction. By defining the leaky scanning score, we find that non-leaky genes are characterized by a Kozak context with a particular enrichment of a cytosine at the –1 and +5 positions and shorter 5' UTR mRNAs. Inhibition by i14G1-12 leads to broad enhancement of leaky scanning, impairment of the 48S ribosome, and an overall decrease in ribosome occupancy at initiation sites. The application of Sel-TCP-seq to isolate 48S complexes associated with eIF1, eIF4G1, eIF3, and eIF2 unveils the remarkable vulnerability of the eIF3-scanning ribosome complex to eIF4G1-eIF1 inhibition and demonstrates the *in vivo* dynamics of the 48S complex during initiation. Specifically, our analysis of the initiation site footprints reveals two major 48S conformations; the first, enriched with eIF2, eIF1, and eIF4G1, marks early initiation, while the other has reduced eIF2 and eIF4G1 occupancy and exhibits an extended coverage of the mRNA by eIF3, suggesting a late or alternative initiation complex. These findings highlight the general outline of scanning and AUG recognition and provide evidence of conformational rearrangements of the 48S during initiation *in vivo*.

RESULTS

The implications of uORFs and eIF1-eIF4G1 inhibition on the scanning 48S

Recently, we reported the identification of inhibitors of the eIF1-eIF4G1 interaction and their respective functions (i14G1s). These small molecules mimic the effects of eIF1 and eIF4G1 knockdown on translation initiation of model mRNAs bearing very short and very long 5' UTRs. They also caused significant changes in translation, as determined by polysome and ribosome profiling.²³ However, as eIF1-eIF4G1 interaction is important for scanning and start codon recognition,^{18,21–23} we were interested in analyzing the role of these factors in these steps. For this purpose, we set out to determine the footprints from 48S ribosomes using TCP-seq of cells treated with the eIF1-eIF4G1 inhibitor i14G1-12 (Figure 1A). We adapted the protocol

from Wagner et al. 2020³¹ with some modifications (STAR Methods). We generated three biological replicates of HEK293T cells treated with DMSO or with 5 μ M i14G1-12 for 1 h, followed by 48S-TCP-seq analysis (Figure 1B). The reproducibility of the replicates from the TCP-seq libraries was high (Figures S1A and S1B). Under these conditions, the presence of eIFs (eIF1, eIF1A, eIF3c, and eIF4G1) in the 40/48S fractions was maintained (Figures S1C and S1D), consistent with the notion that i14G1-12 is an allosteric inhibitor.²³ The length distribution of the reads from the TCP-seq spanned from 17 nt to 80 nt, with a vast majority being shorter than 40 nt (Figure S1E). As expected, most of the rRNA reads originate from the 18S (Figure S1F). Interestingly, upon i14G1-12 inhibition, the iMet tRNA was the most depleted tRNA (Figure 1C). Moreover, the number of reads mapping to mRNAs was significantly decreased by i14G1-12 treatment in the three replicates relative to DMSO (Figure 1D).

We then assessed the distribution of the reads on the mRNA. Most of the reads mapped to the 5' UTR as expected, but we also observed reads on the coding (CDS) and 3' UTR (Figure S1G). We then specifically analyzed the distribution of reads on the 5' UTR using metagene analysis (Figure 1E). As reported by others, the reads accumulated toward the start codon, suggesting that the ribosome gains stability upon scanning or slows down due to queuing.^{31,32} The accumulation of reads from 5' to 3' occurs in two phases, in which the first accumulates faster (Figure 1E). The i14G1-12 caused a progressive reduction of the number of reads on the leader sequence as well as on the start codon relative to DMSO. We performed an analysis of the differentially scanned genes and found 624 downregulated genes (Figure 1F, Table S1) and only 10 upregulated genes. To elucidate how uORFs, which do not overlap the CDS, affect the scanning 48S, we compared genes with experimentally validated uORFs (derived from translation initiation site profiling and ribosome profiling²³) and those lacking uORFs. As expected, uORF genes have a much longer 5' leader than genes without uORFs (Figure S1H). Remarkably, in the absence of uORFs, the slope of the accumulating 48S near the start was further enhanced, while that of uORF genes was dramatically reduced (Figure 1G, left), suggesting that uORFs decreased ribosome queuing near the start. The i14G1-12 effect is negatively correlated with the ribosome occupancy for genes with or without uORFs (Figure 1G, right). These observations reveal the impact of uORFs on the queuing ribosomes and indicate that i14G1-12 treatment leads to destabilization of the scanning 48S.

Global analysis of leaky scanning and the critical role of eIF4G1-eIF1 interaction in its restraining

Our data revealed that the reads did not exclusively map to the 5' UTR but also to the CDS and the 3' UTR (Figure S1E). It has been proposed previously that reads on the CDS originate from elongating ribosomes that dissociate during cross-linking, leaving the 40S subunit on the mRNA.²⁸ On the other hand, the abundance of these reads was significantly increased by i14G1-12 (Figures 2A and S1E). Considering the contribution of eIF1 and eIF4G1 in the AUG recognition,^{14,18,22,23} another possibility is that at least part of the reads mapping to the CDS originates from the ribosome bypassing the start codon. To test this

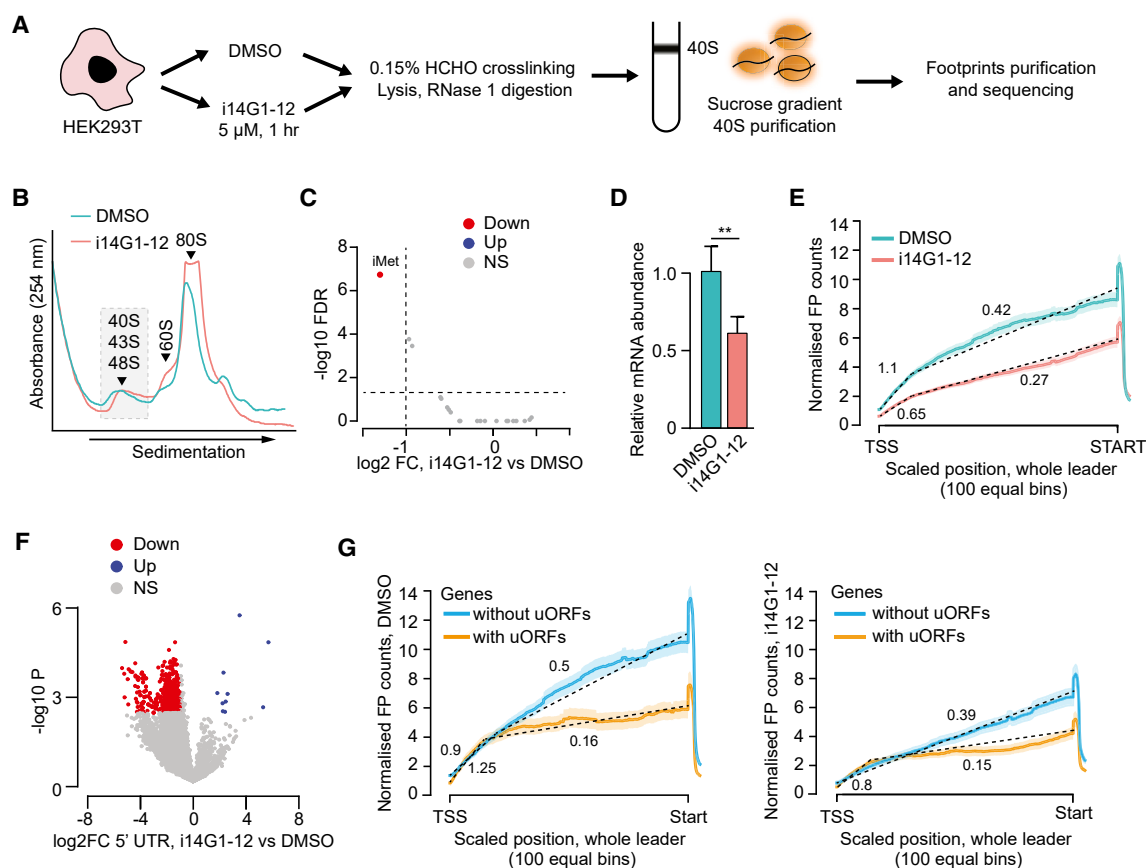


Figure 1. The impact of eIF1-eIF4G1 inhibition by i14G1-12 and uORFs on the 48S

(A) A scheme of the method used. HEK293T cells were treated with DMSO or i14G1-12 for 1 h. The cells were cross-linked on ice with 0.15% formaldehyde before lysis. Cell extracts were digested with RNase I. 40S particles were purified through ultracentrifugation on a sucrose gradient. Protected RNA footprints (FPs) were extracted and processed through library preparation.

(B) Polysome profile of extracts from cells treated or not treated with i14G1-12 and processed through the protocol described in (A). The peaks of 40S, 60S, and 80S are indicated. The area that is highlighted corresponds to the 40S fractions that were used for RNA extraction and library preparation.

(C) Volcano plot showing the log₂ fold change (log₂FC) of tRNA abundance in i14G1-12 over DMSO. tRNAs that are significantly reduced (log₂FC < -1, false discovery rate [FDR] < 0.05) are highlighted in red.

(D) A histogram depicting the relative number of reads that mapped to messenger RNA in the DMSO or i14G1-12 libraries. The mean is reported $\pm$ SD ($n = 3$); asterisks denote a statistically significant difference obtained from a t test between two groups: ** $p < 0.01$.

(E) Metagene analysis of the normalized 40S FP counts on the leader region. Regression lines with slopes are indicated for the two observed phases.

(F) Volcano plot of the Voom-Limma output showing the log₂ fold change of differential FP counts on the leader (5' UTR) in i14G1-12 vs. DMSO. mRNAs that are significantly downregulated (log₂FC < -1, FDR < 0.05) or upregulated (log₂FC > 1, FDR < 0.05) are highlighted in red and blue, respectively.

(G) Metagene analyses of the normalized 40S FP counts on the leader region for mRNAs with or without uORFs under the DMSO (left) or i14G1-12 (right) condition. Regression lines with slopes are indicated for the two observed phases.

See also [Figure S1](#).

hypothesis, we first determined the 3-nt periodicity of the CDS reads based on their 5' and 3' ends and found no enrichment of a specific ORF ([Figure S2A](#)), consistent with the notion that at least part of these complexes do not originate from translating ribosomes. We then defined a leaky scanning score (LS) for each mRNA ([Figure 2B](#)). The score is the log₂ transformation of the ratio between the reads per million per kilobase (RPKM) number of footprints on the CDS up to 40 codons and the number of footprints originating from initiating ribosomes. A positive score suggests that the mRNA has more reads on the CDS than on the start codon ([Table S1](#)). This analysis revealed that most of the transcripts are non-leaky (the score is < 0), but the overall leaky

score was increased by i14G1-12 ([Figure 2C](#)), in line with the metagene analysis. Upon discrimination between two populations of mRNAs either with a strong negative score (LS < -3, defined as non-leaky) or with a strong positive score (LS > 3, defined as leaky), we observe that, under i14G1-12 conditions, the number of genes with a significantly positive score is dramatically increased ([Figure 2D](#)). As the context predetermines whether the translation machinery will bypass a start codon, we looked at the sequence around the annotated start codon of mRNAs with strong positive or strong negative scores. In accordance with the leaky scanning hypothesis, genes with negative scores (non-leaky) had an overall stronger Kozak

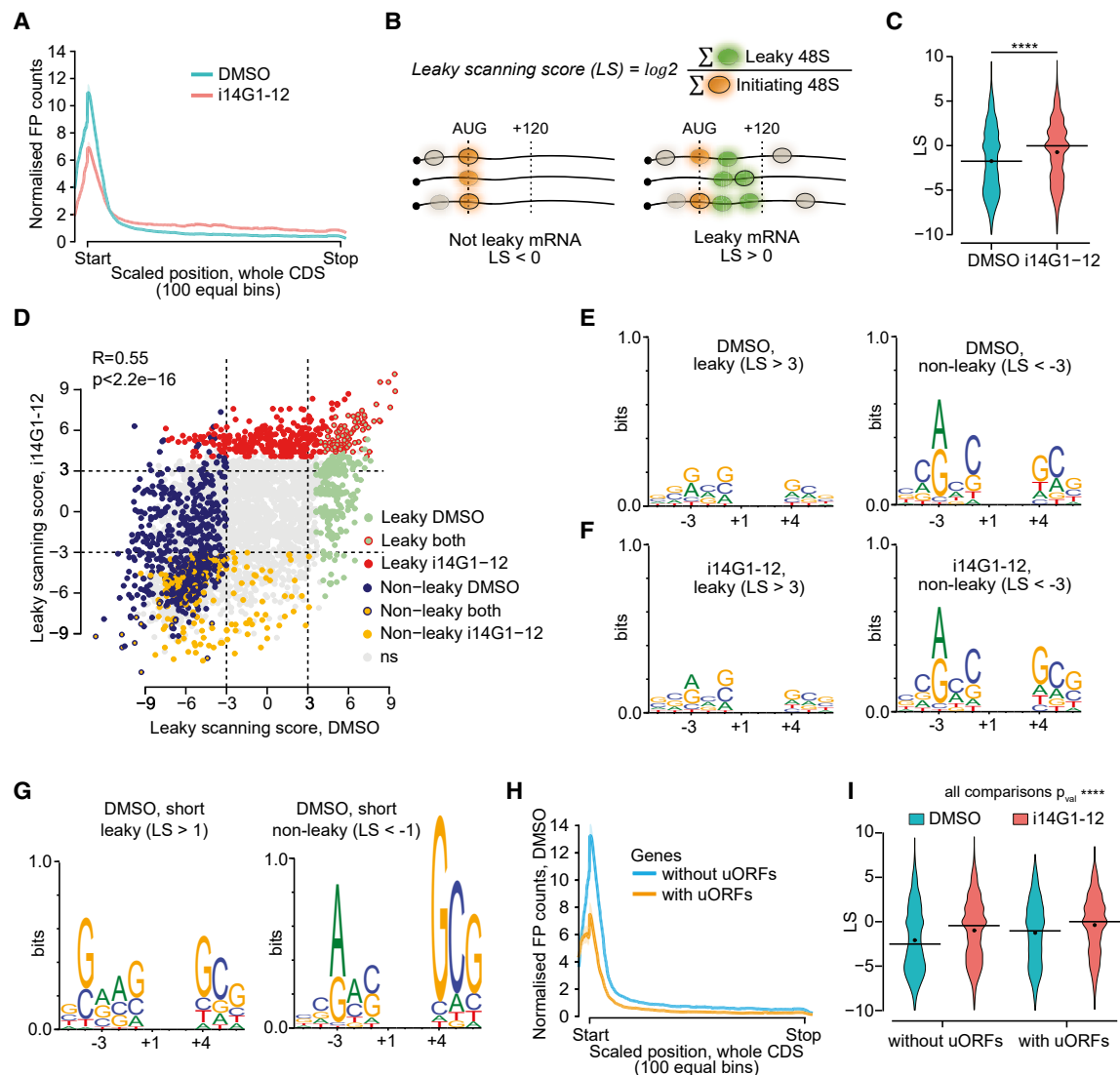


Figure 2. Definition and analysis of leaky scanning and the impact of eIF4G1-eIF1 inhibition

(A) Metagenome analysis of the normalized 40S FP counts on the CDS region.
 (B) Scheme of the leaky scanning score (LS) definition. LS is the log₂ transformation of the ratio between FPs originating from leaky 48S (green ribosome) and initiating 48S (orange ribosome). FPs were defined according to their 3' end distance to the start codon. An initiating 48S has a 3' FP end positioned between 18 and 28 nt from the start codon. A leaky 48S has 3' FPs that are at least distant by 30 nt from the start codon.
 (C) Violin plot showing the LS distribution of mRNAs under the DMSO or i14G1-12 condition. The mean (•) and median (–) are indicated. Asterisks denote a statistically significant difference obtained from a t test between the two groups: *****p* < 0.0001.
 (D) Scatterplot representing the LS of mRNAs in i14G1-12 versus DMSO. mRNAs that are significantly not leaky (LS < –3) or leaky (LS > 3) are highlighted.
 (E and F) Web logos of the motif sequence around the start codon of genes that are strongly leaky (left) or strongly not leaky (right) in DMSO (E) or i14G1-12 (F) treatments.
 (G) Web logos of the motif sequence around the start codon of leaky genes (left) or not leaky ones (right) with a short leader sequence (<30 nt) in DMSO.
 (H) Metagenome analyses of the normalized 40S FP counts on the CDS region for mRNAs containing or not containing uORFs in DMSO.
 (I) Violin plot showing the LS distribution of mRNAs under the DMSO or i14G1-12 condition with and without uORFs. The mean (•) and median (–) are indicated. Asterisks denote a statistically significant difference obtained from a t test between two groups: *****p* < 0.0001.
 See also Figure S2.

context than genes with positive scores (Figures 2E and 2F) and a particular enrichment of cytosine at –1 and +5 positions. In i14G1-12 treated samples, the context of the non-leaky mRNAs is very similar, albeit with some enhancement of the +4G position. Additionally, we noticed that leaky genes have longer 5'

UTRs, and i14G1-12 treatment enhances the LS of shorter 5' UTRs (Figure S2B). This result supports the hypothesis that the reads originating from the CDS region are not only the results of 80S ribosome dissociation but are mostly the consequence of leaky scanning. Gene Ontology term analysis demonstrated

that the non-leaky population was significantly enriched in genes involved in key functions such as metabolism, cell cycle, splicing, and translation (Figure S2C). However, the leaky genes were significantly less enriched in specific pathways (Figures S2C), suggesting that housekeeping genes are resistant to leaky scanning.

AUG codons placed proximal to the 5' cap are known to be susceptible to leaky scanning. We took advantage of our leaky scanning data to examine how start codons positioned up to 30 nt from the 5' cap perform. The context of non-leaky genes was highly enriched in purine at -3 and GCs at positions $+4$ to $+7$ (Figure 2G, right). The frequencies of the $+4$ G and $+5$ C were remarkably higher than those of non-leaky genes with longer leader sequences (Figure 2E vs. Figure 2G). The nucleotide enrichment of the short-leader non-leaky mRNA is reminiscent of the TISU (translation initiator of short 5' UTR) element, of which positions $+4$, $+5$, and $+6$ have been shown to prevent leaky scanning.^{33–35} Interestingly, leaky mRNAs with short leaders bore a stronger Kozak context than those with longer leaders, as they were enriched in $+4$ G (Figure 2G, left). This finding is consistent with the notion that initiation from a short leader requires a more specific motif sequence than on longer leaders.

We also analyzed the association between uORFs and leaky scanning (Figures 2H and 2I) and found that not only is the initiation reduced, but also the LS is enhanced in the presence of uORFs. Upon i14G1-12 exposure, the difference between genes with or without uORFs is largely diminished (Figure 2I). To understand the underlying basis for LS enhancement in uORF-containing mRNAs, we compared their AUG context. The results revealed that uORFs are genetically linked to a weaker AUG context with a particular underrepresentation of a C at position -1 and G at $+4$, features that can drive leaky scanning (Figure S2D).

Altogether, these results demonstrate that 48S reads on the CDS region are derived from ribosomes bypassing the start codon and confirm the importance of a specific AUG context in leaky scanning. Inhibition of the eIF4G1-eIF1 interaction impairs AUG recognition and promotes leaky scanning.

Differential patterns of the 48S scanning ribosome associated with eIF1, eIF4G1, eIF3, and eIF2 α

To assess the impact of the eIF1 and eIF4G1 on the pattern of the scanning 48S ribosome and the implication of their inhibition by i14G1-12, we employed Sel-TCP-seq. In this method, the complex of the 48S with an eIF is immunoprecipitated with specific antibodies, and the linked mRNA is subjected to deep sequencing (Figure 3A). Due to a lack of appropriate antibodies against eIF1, anti-hemagglutinin (HA) antibodies were employed for cells transiently transfected with HA-eIF1. For eIF4G1, a commercial antibody, previously validated for this approach, was used. We also examined, with this method, eIF3c, a subunit of eIF3 that directly binds both eIF1 and eIF4G1,^{5,36} as well as eIF2 α using anti-HA antibodies for HA-eIF2 α -transfected cells. The reproducibility of the replicates was high (Figure S3A). After mapping the reads, we observed that, in all of the Sel-TCP-seq experiments, the most enriched rRNA population is 18S (Figure S3B), and the most abundant tRNA is iMet (depleted upon

i14G1-12 treatment) (Figure S3C), confirming the isolation of the 48S ribosome. The read length distribution shows that, in all Sel-TCP-seq samples, there are two major peaks, one between 15 and 25 nt and the second between 30 and 40 nt (Figure S3D). Following the i14G1-12 treatment, the 30- to 40-nt peak is reduced, while the 15- to 25-nt peak is elevated in Sel-eIF1 and eIF3; the opposite is seen with Sel-eIF4G1. No difference was found with Sel-eIF2 α by the treatment (Figure S3D).

Using metagene analysis, most of the reads were mapped to the 5' UTR (Figure 3B). The profiles of the 5' UTR read distribution of the Sel-eIF4G1, eIF3, and eIF2 α are characterized by an accumulation of reads toward the start site, as seen for the total TCP-seq. In contrast, the pattern of the Sel-eIF1 5' UTR reads is distinct, displaying no accumulation around the start site. This observation is consistent with the *in vitro* release of eIF1 upon AUG recognition.^{37,38} As expected, the treatment with i14G1-12 caused an overall reduction in the 5' UTR reads in Sel-eIF1 and eIF4G1. Remarkably, the inhibitory effect of i14G1-12 is the greatest for Sel-eIF3, while for Sel-eIF2 α , it is minimal. We next examined genes with and without uORFs and found that, for Sel-eIF4G1, eIF3, and eIF2 α , those with uORFs exhibit a similar shallow slope as the total TCP-seq, albeit with a noticeable accumulation at the start. On the other hand, 5' UTRs containing uORFs tend to release eIF1 toward the start site (Figure 3C). The analysis of the LS demonstrates that, as with the total TCP-seq, it was increased by i14G1-12 for Sel-eIF1, eIF4G1, and eIF3, but that of eIF2 α was unaffected (Figure 3D), consistent with the lack of the significant 3 nt (Figure S3E). Altogether, these findings strongly suggest that i14G1-12 targets eIF1 and eIF4G1 in the context of eIF3.

Initiation site footprints unveil AUG recognition modes and the 48S initiation complex dynamics

From the metagene analysis (Figures 1E and 2A), most of the reads accumulate at the start codon. We assessed the distribution of the reads that span the AUG by their exact 5' and 3' boundaries and their length (Figures 4A and S4A). We noticed that the ends of the reads clustered at specific positions, which are all affected by i14G1-12. In general, most of the 5' ends are located upstream of the start codon at variable positions from -12 to -50 , depending on the read length (Figures 4A and 4B, left). As proposed previously,^{31,39} this observation suggests that the 48S composition is heterogeneous at the 5' end. On the other hand, most of the 3' ends are distributed in a narrow window of a few nucleotides downstream of the start codon from $+24$ to $+26$ (Figures 4A and 4B, right). According to their length and end positions, we find reads that span the AUG (A population; Figure 4A, scheme) and reads that are cleaved at $-1/+1$ (B and C populations; Figure 4A, scheme). As they surround the start codon, we defined these populations as footprints from initiating ribosomes. *In vitro* data have shown that the P site surrounding nucleotides becomes accessible to the nuclease RelE following the initiation;^{40,41} therefore, the B and C read populations are most likely cleavage products from initiating ribosome footprints at $-1/+1$ by RNase I. In line with this notion, there is a significant overlap between B and C as well as A populations (Figure 4C),

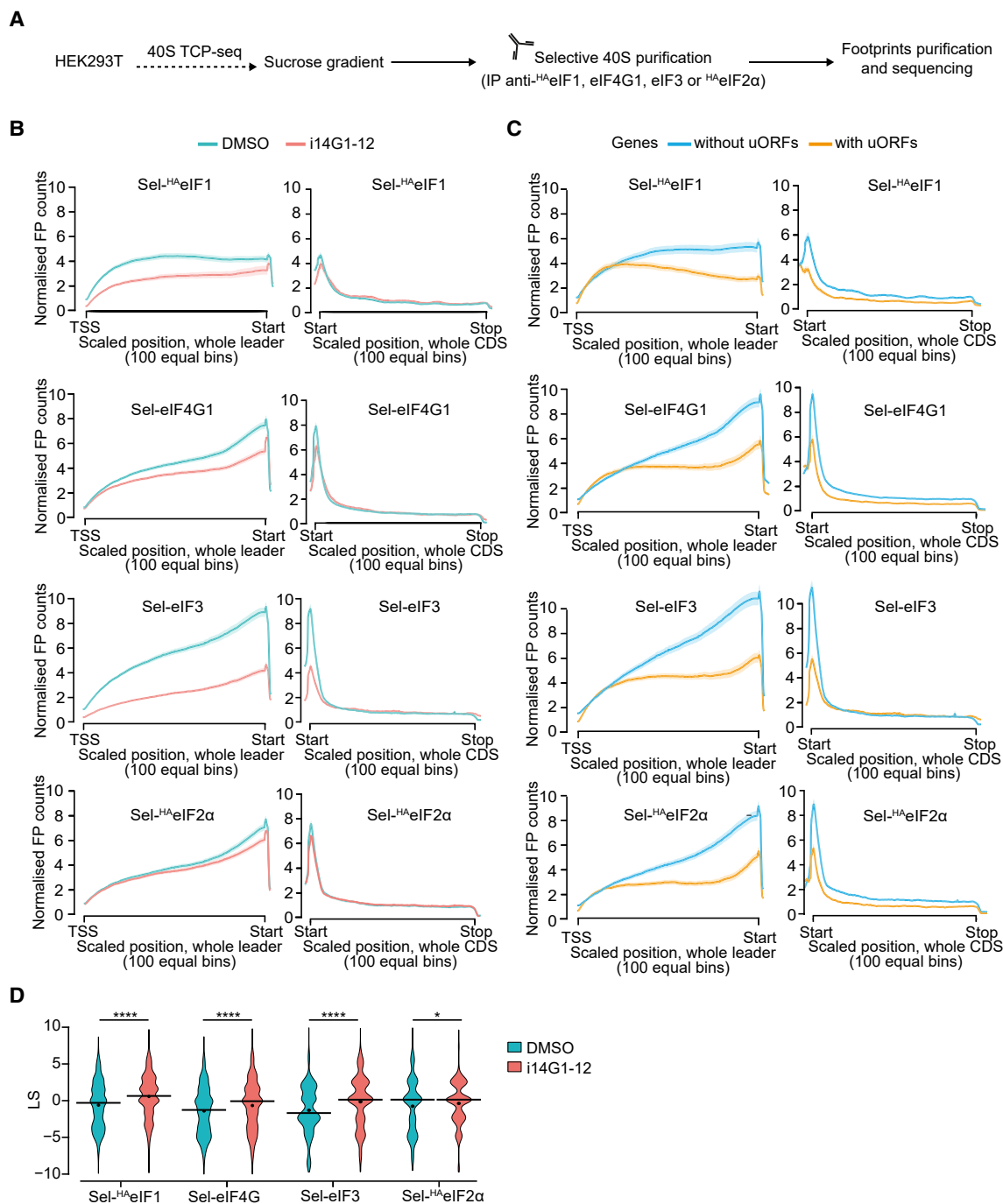


Figure 3. Analysis of scanning 48S ribosomes in association with eIF1, eIF4G1, eIF3, and eIF2α through Sel-TCP-seq

(A) A scheme of the Sel-TCP-seq that was used. The indicated steps described here are subsequent to the 40S TCP-seq protocol depicted in Figure 1A. 40S complexes assembled with a specific eIF were immunoprecipitated before FP extraction. For eIF2α and eIF1, HEK293T cells were transiently transfected with vectors encoding for HA-tagged proteins.

(B) Metagenesis analyses of the normalized 40S FP count on the leader (left) or CDS (right) region from the different Sel-TCP-seq for all genes in DMSO and i14G1-12.

(C) Metagenesis analyses as in (B) for genes with or without uORFs in DMSO.

(D) Violin plots showing the LS distribution of mRNAs under the DMSO or i14G1-12 condition after Sel-TCP-seq. The mean (•) and median (–) are indicated. Asterisks denote statistically significant differences obtained from a t test between the two groups: * $p < 0.05$ and **** $p < 0.0001$, respectively.

See also Figure S3.

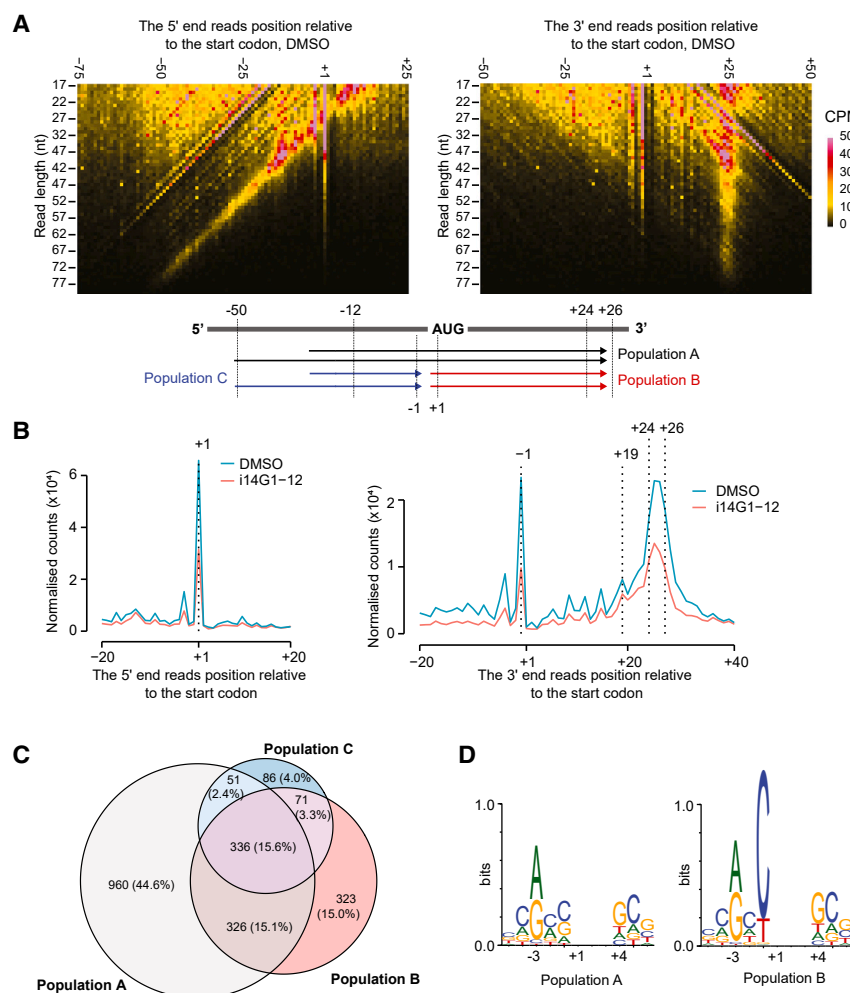


Figure 4. Analysis of 48S initiation site FPs

(A) Matrices of the read length versus the distance of 5' end (left) or 3' end (right) read positions relative to the start codon in DMSO. The sum of normalized counts per million of the three replicates for each position and length is displayed. A scheme that illustrates reads labeled as population A, B, or C is shown below the matrices. Population A read ends span between +19 and +26 from the start codon with length ranging from 30 to 80 nt. Populations B and C are reads starting at +1 or ending at -1, respectively.

(B) Plots of the normalized 5' end read counts (left) or 3' end read counts (right) around the start codon in DMSO and i14G1-12.

(C) Venn diagram of genes enriched in population A, B, or C reads.

(D) Web logos representing the motif sequence around the start codon of transcripts from populations A (left) and B (right). See also Figure S4.

suggesting that they originate from cleavage within the initiating 48S. According to the read counts, population B was more retained than population C, suggesting that population C is less protected (Figure 4C).

To elucidate the underlying basis for the -1/+1 cleavage, we compared the surrounding AUG sequence of genes from A and B populations (Figure 4D). Both A and B bear a strong Kozak context, as evident from the enrichment of A/G at position -3 and G at +4. However, a C at position -1 clearly distinguishes population B from A. While the possibility that this cytosine is a preferred RNase I cleavage site cannot be ruled out, the observation that other heightened cleavage sites, such as +24-26 (Figure S4B), are not enriched with a C suggests that the -1 cytosine has a specific impact on the 48S conformation by exposing it to RNase I cutting.

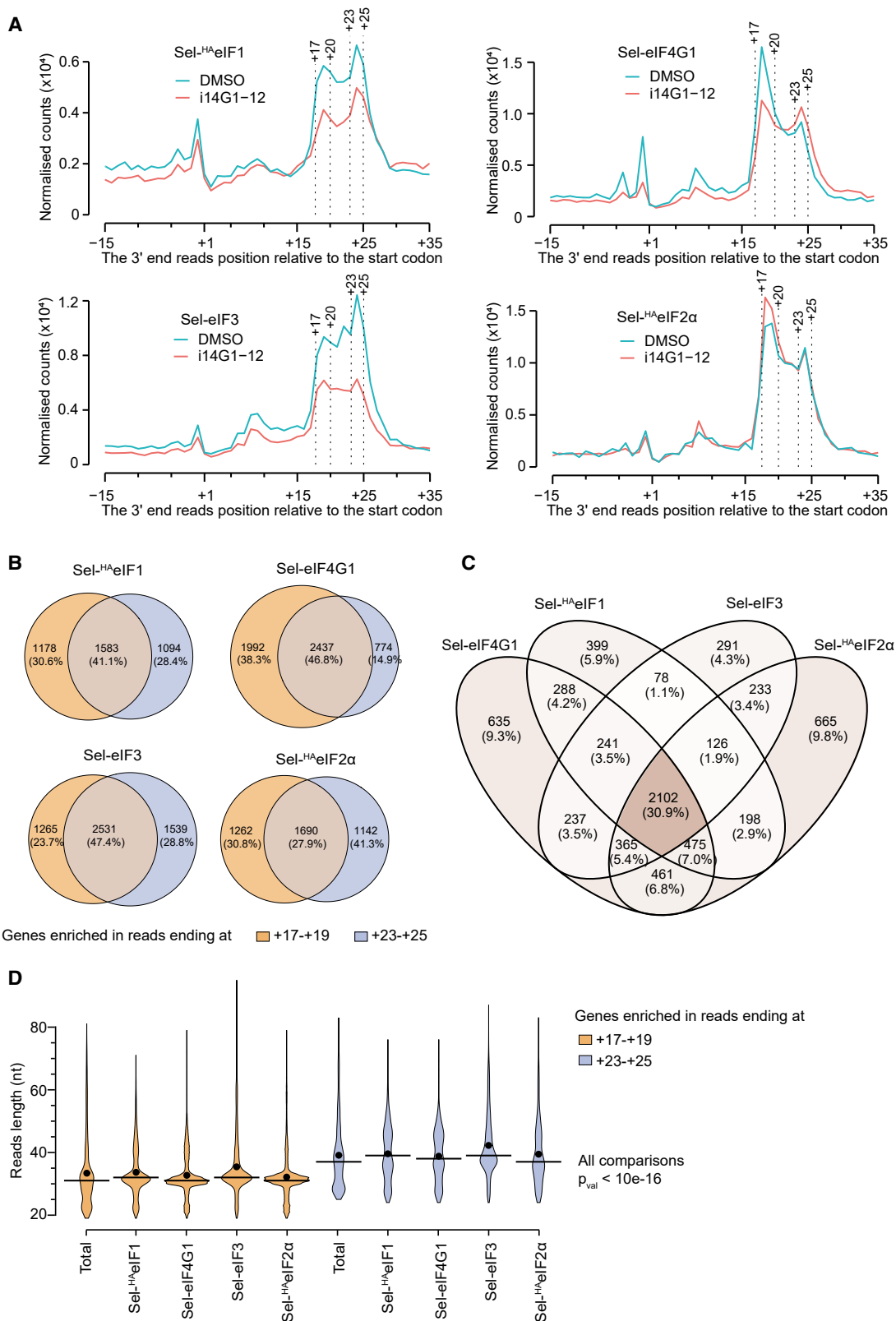
According to the distribution of the footprints around the start codon, most of the 3' ends of populations A and B are positioned between +24 and +26 from the start codon (Figure 4B, right). As these reads originate from initiating ribosomes, this finding suggests that the 48S covers the mRNA up to +27 from the P site. However, in many previous *in vitro* assays, such as toeprinting, the 48S and the 80S cover the mRNA up to +15 to +18 nt from

the P site. This extension is also seen by analyzing reported and publicly available TCP-seq data of mammalian and yeast cells,^{29,31,32} including HEK293T cells (Figure S4C), even though the previous experiments used somewhat different protocols, which may account for the differences in distribution. We compared the AUG initiation context of genes that are enriched with the +24 to +26 footprint to those lacking it and found that the +24 to +26 extension is highly prevalent in genes with a strong Kozak context (Figure S4D).

To better characterize the source of the footprint extension, we analyzed the Sel-TCP-seq data.

Remarkably, with all Sel-eIFs, we observed a significant enrichment of the +17 to +19 3' end reads in comparison to the total TCP-seq, where these reads constitute a minor population (Figure 4B, right, vs; Figure 5A). The +17 to +19 peak is particularly enriched in the Sel-eIF1, eIF4G1, and eIF2 α . Interestingly, the second peak of the +23 to +25 extension is more abundant in Sel-eIF3 and eIF1, while eIF2 α and eIF4G1 are relatively reduced (Figure 5A). This finding suggests that the extended +23 to +25 48S footprint is generated by eIF3. This notion is supported by analyzing published eIF3 Sel-TCP-seq data from *S. cerevisiae* and HeLa cells^{29,30} showing similar enrichment of these positions (Figure S5A). We examined the effect of i14G1-12 on the read distribution and observed a substantial reduction for both peaks with Sel-eIF1, eIF4G1, and eIF3 but not with eIF2 α . Interestingly, the effect is differential and aligns with the targeted protein (Figure 5A). Altogether, these findings support the idea that +17 to +19 and +23 to +25 represent two 48S conformations that differentially engaged with the tested eIFs.

Next, we determined the extent of overlap between the two major peaks to determine whether they represent two different conformations of the same transcripts or two distinct initiation complexes and found a significant albeit not full intersection



(legend on next page)

(Figure 5B). Notably, the number of reads (Figure 5A) and genes (Figure 5B) co-immunoprecipitated with eIF1 are much smaller relative to the other tested eIFs, consistent with the conclusion of the metagenome analysis (Figure 3B) that part of eIF1 is released upon initiation. Likewise, the overlap of enriched genes from the four Sel-TCP-seq in both initiation positions revealed a substantial number of common genes (Figures 5C and S5B). Genes that are common to all of the Sel-eIFs are enriched with a better AUG context, further confirming the enhanced stability conferred by the context (Figure S5C).

To gain further insights into these initiation complex confirmations, we determined the length of the footprints ending either at +17 to +19 or the extended +23 to 25. In the total and Sel-eIF1, eIF4G1, eIF3, and eIF2 α , the mean length of the +17 to +19 is 32 nt, reflecting upstream protection at around -12 (Figure 5D), consistent with 5' end analysis (Figure S5D). For the extended +23 to 25, the mean length and distribution of Sel-eIF3 are greater than 42, while that for the others is 37, reflecting 5' end protection of > -15 or -12, respectively. These data also confirm that eIF3 is responsible for the extension of the footprint of the 48S ribosome.

DISCUSSION

In this study, we dissected the scanning and AUG recognition phases of translation initiation by analyzing the 48S-mRNA complexes through TCP-seq and Sel-TCP-seq. Additionally, we examined the consequences of perturbing eIF1-eIF4G1 function using the small-molecule inhibitor i14G1-12. Our findings provide several important insights into translation initiation *in vivo*. First, we demonstrate that the presence of validated uORFs significantly diminishes scanning ribosome density and queuing near the start site. Second, we constructed, for the first time, a landscape of leaky scanning of endogenous mRNAs and identified the features driving it. Third, we demonstrate the importance of eIF4G1-eIF1 in scanning and initiation and its central role in restricting leaky scanning. We further identified two major conformations of the initiating 48S that differentially engage with eIF1, eIF4G1, eIF2 α , and eIF3. Finally, the collective data of 48S-eIFs complexes strongly suggest that the inhibitory effect of i14G1-12 on eIF1-eIF4G1 is closely linked to their association with eIF3.

We have established that TCP-seq reads mapping to the CDS are mainly the results of leaky scanning. Upon defining an LS for each gene, we observed a negative correlation between leakiness and the strength of the AUG context. As expected, the important determinant that influences the fidelity of the translation initiation is the Kozak AUG context. Additionally, we found that a C at positions -1 and +5 for a long 5' UTR and CG at +5

and +6 for a very short 5' UTR are important features for resistance to leaky scanning. In the cryoelectron microscopy structure of the 48S,⁴² the -1 C position is relatively isolated, with no nearby ribosomal protein or initiation factor. This may explain the enhanced accessibility of this position to RNase I digestion. The +5 and +6 CG of the very short 5' UTR were previously studied in the context of the TISU element and have been shown to be critical for the resistance to leaky scanning,^{33,35} and these positions were cross-linked to RPS3 and RPS10.³⁵ The importance of RPS3 mRNA binding for the translation of an extremely short 5' UTR has been demonstrated most recently.⁴³ Thus, these sequence features are very important to maintain the stability of the 48S following recognition of the start codon.

The examination of the importance of eIF4G1-eIF1 during scanning and initiation at a global scale provides evidence for the central role of this complex in preserving the integrity of the 48S, consistent with the original study in yeast on eIF4G1-eIF1.¹⁸ The inhibition of eIF4G1-eIF1 led to a decrease in ribosome occupancy on the 5' UTR, which is likely a result of less scanning and queuing ribosomes. This is supported by the diminution of initiating ribosomes on the start codon. In line with previous genetic and molecular studies,^{18,22} the eIF1-eIF4G1 complex plays an important role in appropriate AUG recognition and prevention of leaky scanning, as, upon short-term i14G1-12 treatment, leaky scanning is markedly and globally elevated, with the exception of mRNAs bearing a strong Kozak context. Altogether, the inhibition by i14G1-12 provides strong support to the central roles of eIF1 and eIF4G1 in maintaining 48S ribosome integrity during scanning, facilitating initiation, and restricting leaky scanning.

Employment of the powerful Sel-TCP-seq to assess the impact of eIF1, eIF4G1, eIF3c, and eIF2 α on the scanning and initiating 48S ribosome and the consequences of eIF1-eIF4G1 inhibition by i14G1-12 provided interesting findings. Specifically, the 5' UTR read distribution of Sel-eIF1 displayed a unique profile with no accumulation around the start site, consistent with a significant release of eIF1 upon AUG recognition. Yet, in a significant number of genes, eIF1 is maintained as part of the initiation complex (Figure 5A), suggesting the existence of 48S initiation complexes with and without eIF1 (Figure 6). Treatment with i14G1-12 led to an overall reduction in 5' UTR reads for Sel-eIF1, eIF4G1, and Sel-eIF3 but not Sel-eIF2 α . Interestingly, the i14G1-12 inhibitory effect on scanning was the greatest for the Sel-eIF3, suggesting that the inhibitor targets eIF1 and eIF4G1 in the context of eIF3c. Indeed, both eIF1 and eIF4G1 interact directly with eIF3c.^{22,44} This conclusion is also supported by the differential effect of i14G1-12 on the two 48S initiation conformations (see below). The Sel-TCP-seq also showed that the presence of uORFs influenced the read distribution in Sel-eIF1, eIF4G1, and eIF2 α and promoted leaky

Figure 5. Analysis of initiation-site FPs of the Sel-eIFs-TCP-seq and definition of 48S initiation complex dynamics

- (A) Plots showing the normalized 3' end FP counts around the start codon in DMSO and i14G1-12 from the Sel-TCP-seq.
(B) Venn diagram illustrating the overlap between genes enriched in FPs with 3' ends spanning between +17 and +20 or between +23 and +25 in each of the four Sel-TCP-seq.
(C) Venn diagram illustrating the overlap of the mRNAs enriched in reads with 3' ends between +17 and +25 in the four Sel-TCP-seq.
(D) Violin plots showing the length in nucleotides of the FP with 3' ends within the +17 to +20 or +23 to +25 range. Included reads are only those that overlap the start codon. Total refers to the DMSO TCP-seq (Figure 4B). The mean (•) and median (–) are indicated.

See also Figure S5.

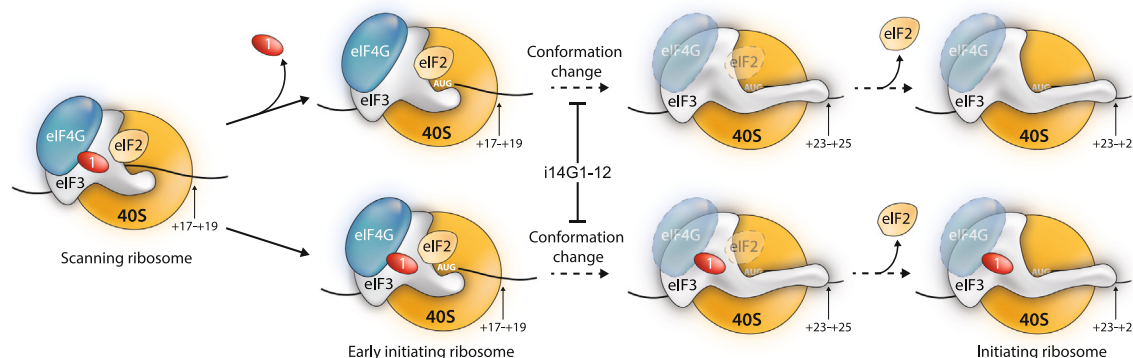


Figure 6. A model describing the different 48S initiation complexes

The numbers indicate the 3' edge of the 48S complex on the mRNA relative to the start codon. The scanning ribosome is engaged with all eIFs. Upon AUG recognition, eIF1 is released from a fraction of mRNAs, resulting in two species of early initiation complexes with and without eIF1. Subsequently, or in parallel, the complex is re-arranged by eIF3, resulting in an extension of mRNA occupancy up to position +25. At this stage, eIF2 and eIF4G1 are loosely associated. Upon initiation, eIF2 is released. i14G-12 inhibits the rearrangements, leading to leaky scanning.

scanning, most likely due to a genetic linkage with the weaker AUG context (Figure S2C).

Analyzing the initiation site footprints from the TCP-seq and Sel-TCP data enabled us to assign modes of initiating ribosomes based on the boundaries of the 48S protection from RNase I digestion. Our findings provide the first *in vivo* evidence for two major intermediate phases during initiation (Figures 4 and 5). Specifically, the analysis of the 5' and 3' edges of the initiation site footprints derived from the total and the eIF immune-precipitated complexes revealed that one significant conformation of the 48S that protects the mRNA up to the +17 to +19 positions is enriched predominantly with eIF2 α and eIF4G1. As eIF2 α is expected to be released from the 48S upon AUG recognition, the presence of high levels of eIF2 α at the +17 to +19 positions suggests an early initiation phase (Figure 6). In the second conformation, the footprint is extended to +23 to +25 and is particularly enriched with eIF3 $\pm$ eIF1, while eIF2 α and eIF4G1 are reduced but not fully released (Figure 6). The eIF3-mediated extension is seen from both sides of the AUG and is a major determinant of the second major 48S initiation conformation. We also found evidence for the two 48S forms of initiation in another cell type and organism upon the analysis of published TCP-seq data (Figure S5A), indicating the basic and conserved nature of these dynamic phases. Considering that a significant level of eIF2 α is still present in the eIF3-dependent extended conformation (Figure 5), it remains to be seen whether this configuration specifies a late or alternative initiation phase (Figure 6). Interestingly, i14G1-12 differentially affects the relative amounts of the two conformations, raising the possibility that this transition is important for preventing leaky scanning. Upon completion of the initiation phase by the formation of the 80S, it is expected that a fraction of eIF3 and eIF4G1 remains associated with early elongating ribosomes.^{24–27} This raises the interesting possibility that translation initiation inhibitors that target these factors, such as i14G1-12, may interfere with this process as well.

In summary, our findings highlight the general outline of scanning, AUG recognition, leaky scanning, and the dynamics of the

48S during initiation *in vivo* and how these are controlled by mRNA features and eIFs. Future studies aiming to further explore the 48S rearrangement by structural and functional studies will broaden our understanding of the intricate interplay between the ribosome, eIFs, and the mRNA and the implications of these regulatory mechanisms on cellular processes.

Limitations of the study

While the TCP-seq and Sel-TCP-seq approaches used in this study provided important insights into the scanning and initiation 48S complex, they present several limitations. These include a bias toward well-translated mRNAs and strong AUG context and the inability to identify less abundant or unstable conformation intermediates. The use of exogenously expressed eIF1 and eIF2 α may change the endogenous concentrations of these factors and affect global translation. Finally, analyzing additional eIFs could provide a deeper understanding of the initiation process.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.celrep.2024.114126>.

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AUTHOR CONTRIBUTIONS

Conceptualization, B.W. and R.D.; methodology, B.W. and R.D.; software, B.W.; investigation, B.W.; visualization, B.W.; writing – original draft, B.W. and R.D.; writing – review & editing, B.W. and R.D.; funding acquisition, B.W. and R.D.; supervision, R.D.

DECLARATION OF INTERESTS

The authors declare a patent application for the i14G1–12 used in this study (YEDA-P-015-IL).

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Monoclonal Anti-HA–Agarose antibody produced in mouse	Sigma	Cat# A2095; RRID:AB_257974
Anti-eIF3 p110 (B-6)	Santa Cruz	Cat# sc-74507; RRID:AB_1122487
Anti-eIF4G1	MBL	Cat# RN002P; RRID:AB_1570635
Anti-eIF1A	Abcam	Cat# ab172623; RRID:AB_2811162
Anti-HA	Abcam	Cat# ab91110; RRID:AB_307019
Anti-RPS14	Abcam	Cat# ab174661; RRID:AB_2905612
Chemicals, peptides, and recombinant proteins		
i14G1-12	Thermo scientific	JFD00979SC
Formaldehyde	J.T.Baker	Cat# JTB1715288; CAS Number: 50-00-0
RNase I	Thermoscientific	Cat# EN0601
Deposited data		
40S TCP-seq data under i14G1-12 treatment	This analysis	GEO: GSE226132
Selective 40S TCP-seq data under i14G1-12 treatment	This analysis	GEO: GSE253652
<i>S. cerevisiae</i> and HEK293T TCP-seq data	Wagner et al. ³¹	GEO: GSE139132
<i>S. pombe</i> TCP-seq data	Duncan et al. ³²	ArrayExpress: E-MTAB-12264
HeLa and NIH3T3 TCP-seq data	Bohlen et al. ²⁹	GEO: GSE139391
Experimental models: Cell lines		
Human: HEK293T cells	ATCC	ATCC: CRL-3216
Recombinant DNA		
Pcruz-HA-eIF1	Haimov et al. ²²	N/A
pcDNA3-HA-eIF2 α	N/A	N/A
Software and algorithms		
Cutadapt	Martin et al. ⁴⁵	https://cutadapt.readthedocs.io/en/stable/
Bowtie	Langmead et al. ⁴⁶	https://bowtie-bio.sourceforge.net/index.shtml
HISAT2	Kim et al. ⁴⁷	https://daehwankimlab.github.io/hisat2/
FeatureCounts	Liao et al. ⁴⁸	https://rnah.github.io/bioinfo-notebook/docs/featureCounts.html
deepTools2.0	Ramirez et al. ⁴⁹	https://deeptools.readthedocs.io/en/develop/index.html
GraphPad Prism version 8.4.3	N/A	https://www.graphpad.com/
R Statistical Software version 4.2.0	R Core Team	https://www.r-project.org/
g:Profiler	Kolberg et al. ⁵⁰	https://biit.cs.ut.ee/gprofiler/gost
Samtools	Li et al. ⁵¹	http://www.htslib.org/
Bedtools	Quinlan et al. ⁵²	https://bedtools.readthedocs.io/en/latest/
EdgeR	Robinson et al. ⁵³	https://bioconductor.org/packages/release/bioc/html/edgeR.html
Trimmomatic	Bolger et al. ⁵⁴	http://www.usadellab.org/cms/?page=trimmomatic
WebLogo 3.7.12	Crooks et al. ⁵⁵	https://bioweb.pasteur.fr/packages/pack@weblogo@3.7.4

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Rivka Dikstein (rivka.dikstein@weizmann.ac.il).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- This paper analyzes existing, publicly available data. In addition, TCP-seq and Selective TCP-seq data have been deposited on GEO and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#).
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

HEK293T cells were grown to 90% confluence at 37°C in a humidified incubator supplemented with 5% CO₂. DMEM supplemented with 10% FBS and antibiotics was used as media. The cell line originates from ATCC and has not been further authenticated.

METHOD DETAILS

TCP-seq

40S TCP-seq protocol was adapted from Wagner and co-workers³¹ with some modifications ([Figure 1A](#)). Three biological replicates per treatment were done as follow. HEK293T cells were grown to 90% confluence at 37°C in a humidified incubator supplemented with 5% CO₂. Cells were treated with DMSO or 5 μM i14G1-12 for 1 h. To crosslink the cells, 0.15% formaldehyde was directly added to the media from a 10% solution and the cell plates were then put on ice. Crosslinking reaction was performed for 10 min with slow shaking. The reaction was stopped by the addition of glycine to a final concentration of 37.5 mM and slow agitation for an additional 5 min. Media was aspirated and the plates were washed twice with cold PBS. Cells were scraped in fresh lysis buffer⁵⁶ (100 mM KCl, 20 mM Tris-HCl pH 8, 1.5 mM MgCl₂, 1.5% (v/v) NP-40, 2 mM DTT) supplemented with protease inhibitor and cells were lysed for 20 min in a cold room with slow agitation. Lysates were clarified by centrifugation at 20,000g and RNA amount was quantified by nanodrop. RNase I (ThermoFischer) was added to digest free RNA for 30 min at room temperature. The digestion was stopped by the addition of 5 μL of Super RNase inhibitor (ThermoFischer). Lysates were loaded onto a 5–45% sucrose gradient (100 mM KCl, 20 mM HEPES pH 7, 5 mM MgCl₂, 2 mM DTT) and centrifuged at 39 000 rpm for 2 h and 45 min in an SW41 rotor at 4°C.⁵⁶ Gradients were analyzed through a UV lamp (254 nm) and an Absorbance detector while being fractionated in 250 μL fractions using a Foxy Junior Fraction Collector (Isco). To successfully recover the 48S ribosome, we had to calibrate concentrations of RNase I, as the originally suggested concentration³¹ resulted in a complete loss of the 40/48 peak in the sucrose gradient. We found that decreasing the concentration of RNase I to 0.0125 U per μg of RNA maintained the integrity of a sufficient amount of the 40S as shown in [Figure 1B](#). Similar low RNase I concentrations were previously successfully used for library preparation for ribosome profiling.⁵⁷

Selective TCP-seq

Selective TCP-seq was adapted from the TCP-seq protocol, as described in [Figure 3A](#). For Sel-eIF2α and Sel-eIF1, HEK293T cells were transfected with HA-tagged plasmids. Following sucrose gradient, fractions corresponding to the 40S were pulled and supplemented with 1% Triton X-. The 40S complex was immunoprecipitated with protein A/G Sepharose beads coated with specific antibodies or HA-agarose beads (Sigma). After IP, beads were washed and processed through RNA extraction. Sel-eIF4G1 and Sel-HA-eIF1 were performed in three biological replicates while Sel-eIF3 and Sel-HA-eIF2α were prepared in biological duplicates.

RNA extraction and library preparation

RNA from fractions corresponding to the 40S or from the beads after IP were extracted by the addition of 1 mL of TRIzol. Samples were heated at 65°C for 20 min and vortexed every 5 min. Extraction was continued by the addition of chloroform and then by alcohol precipitation of the aqueous phase. TCP-seq libraries were prepared as previously described by selecting footprints with a length ranging from 16 nt to 85 nt.⁵⁸ The TCP-seq and the Selective TCP-seq libraries were sequenced at the iGE3 Genomic Platform (UNIGE) on a HiSeq 4000 and at the Crown Institute for Genomics (Weizmann Institute of Science) on a NovaSeq 6000, respectively.

Western blot

HEK293T cells were transfected with HA-eIF1 and processed through the TCP-seq protocol under DMSO or i14G1-12 treatment. After fractionation of the sucrose gradient, 40S fractions were pulled. Proteins were precipitated by Methanol-Chloroform and the pooled content was analyzed by western blot with the indicated antibodies.

Mapping

Adapters were trimmed from TCP-seq and sel-TCP-seq libraries with cutadapt⁴⁵ (parameters: -a CTGTAGGCACCATCAAT -m 16) and reads were filtered by quality with TRIMMOMATIC⁵⁴ (parameters: MINLEN:16 TRAILING:15 SLIDINGWINDOW:5:15). Reads were filtered out by successive mapping to rRNAs (UCSC) and to tRNAs (GtrRNA database) with bowtie⁴⁶ and to lncRNAs (Ensembl) with HISAT2.⁴⁷ Unmapped reads were finally aligned to the human MANE transcriptome (Ensemble release 108) with HISAT2 using default parameters. Finally, only primary alignments were kept for the analysis.

For the TCP-seq data from yeast, HeLa and NIH3T3 cells, mapping was performed according to the library preparation method used by the authors.^{29,31,32} For *S. cerevisiae* and *S. pombe*, EnsemblFungi R64-1-1.108 and EnsemblFungi ASM294v2 were used as transcriptome, respectively.

TCP-seq analysis

A custom annotation file of the MANE transcriptome was created from BioMart (Ensembl v108). Genes were counted with FeatureCounts (31) and transcripts were kept if they had a CPM ≥ 10 in all triplicates for either the DMSO or the i14G1-12 treated samples. This left the analysis with 9337 transcripts. The distance between reads and the annotated start codon was determined with Bedtools.⁵² This distance was used in C and R scripts for most of the analysis.

Differential scanning and initiation analysis

Differential initiation analysis was conducted using Limma.⁵³ Reads originating from both the 5' UTR and initiation sites were tallied, normalized by RPKM, and employed as input. The analysis applied a threshold of a log2 fold change of 1, coupled with a false discovery rate (FDR) of 0.05 for significance.

Metagene

Metagene analysis was done with the deepTools package.⁴⁹ Coverage was performed on bins of 1 and normalized using CPM with exact scaling (parameters: -exactScaling -binSize 1 -normalizeUsing CPM). Plots were scaled by mRNA features into 100 equal bins.

Leaky scanning score

To calculate the leaky scanning score, we defined 'initiating' and 'leaky' reads based on the respective distances of their 5' and 3' ends from the annotated start codon. Observing the distribution of reads around the start codon, a read was deemed 'initiating' if its 3' end fell within a window of 18–29 nucleotides downstream of the AUG (Figures 4A and 4B). To include reads that are products of cleavage at the -1/+1 position, 'initiating' reads 5' end can be located up to the +1 position. On the other hand, a read originates from a 'leaky' 48S complex if its 3' end is distant by at least 30 nucleotides from the start codon. 'Leaky' reads were considered only up to the 40th codon (3' ends up to 150 nt) to prevent the inclusion of reads that do not originate from leaky scanning. Within this distance, more than half of the genes in our analysis had at least one AUG codon. Initiating and leaky reads were then aggregated, and the counts were normalized according to the feature length and the library size: 3 nt (1 codon) and 117 nt (39 codons), respectively. Finally, the normalized counts were used as the input for Voom-Limma.⁵³ The logFC output was defined as the leaky scanning score per mRNA, with a false discovery rate (FDR) of 0.1 used as the threshold. The following formula summarizes LS:

$$\text{Leaky Scanning Score} = \log_2 \left(\frac{\sum \text{RPKM}(\text{Leaky 48S 39 codons upstream start})}{\sum \text{RPKM}(\text{Initiating 48S})} \right)$$

QUANTIFICATION AND STATISTICAL ANALYSIS

Student's t-tests were conducted between two groups using GraphPad Prism (Version 8.4.3, GraphPad Prism Software) or R version 4.2.0. A *p*-value lower than 0.05 was considered as significant. Data reported the mean $\pm$ SEM or SD as described in the figure legends.